

The glucose crash is not a normalization artifact of the relative benefit metric

Re-scoring mutualism with a fixed ABSOLUTE benefit (orange) reproduces the L5→L6 crash of the relative metric (blue), even though μ_{alone} jumps at L6 (grey dashed, +35% *Akkermansia* / +42% *Lactobacillus*). Dotted red line = L5→L6 transition.

○ Absolute benefit $\mu_{\text{co}} - \mu_{\text{alone}} > 10^{-3} \text{ h}^{-1}$ △ Relative benefit $\beta = (\mu_{\text{co}} - \mu_{\text{alone}}) / \mu_{\text{alone}}$

